

Appearance-Factored LeWM-lite on Official PushT

Reliable Rerun and Ablation Summary

April 26, 2026

Bottom line. The Stage 2 reliability check favors **Baseline LeWM: 25/400** successes at epoch 50, versus **22/400** for **v1_current** and **20/400** for **v2_app_nuisance_only**. Stage 1 produced useful Appearance-Factored leads, but those leads did not persist after the longer two-seed check.

Architecture and Design Principle

Appearance-Factored LeWM-lite (AF-LeWM-lite) keeps the LeWM planning contract intact. Here, AF means appearance-factored: the added branch factors appearance information into `app_emb`, while planning still uses the dynamics latent `emb`. The action block encoder is inherited from baseline LeWM.

Pixels are encoded by a shared ViT-tiny backbone; the dynamics projection head produces `emb`, action blocks are encoded separately, and the autoregressive predictor rolls the dynamics latent forward under candidate actions. Planning uses terminal latent distance between predicted dynamics and the goal dynamics latent.

The AF extension adds an appearance projection head `app_emb` used only for representation shaping during training. `v1` adds dynamics invariance under appearance augmentation and a cross-covariance penalty between `emb` and `app_emb`. `v2` adds sequence-consistent augmentation parameters, stop-gradient invariance, an appearance nuisance head, and a gradient-reversal nuisance head on the dynamics branch.

Architecture comparison: Baseline LeWM vs. AF-LeWM-lite v1/v2

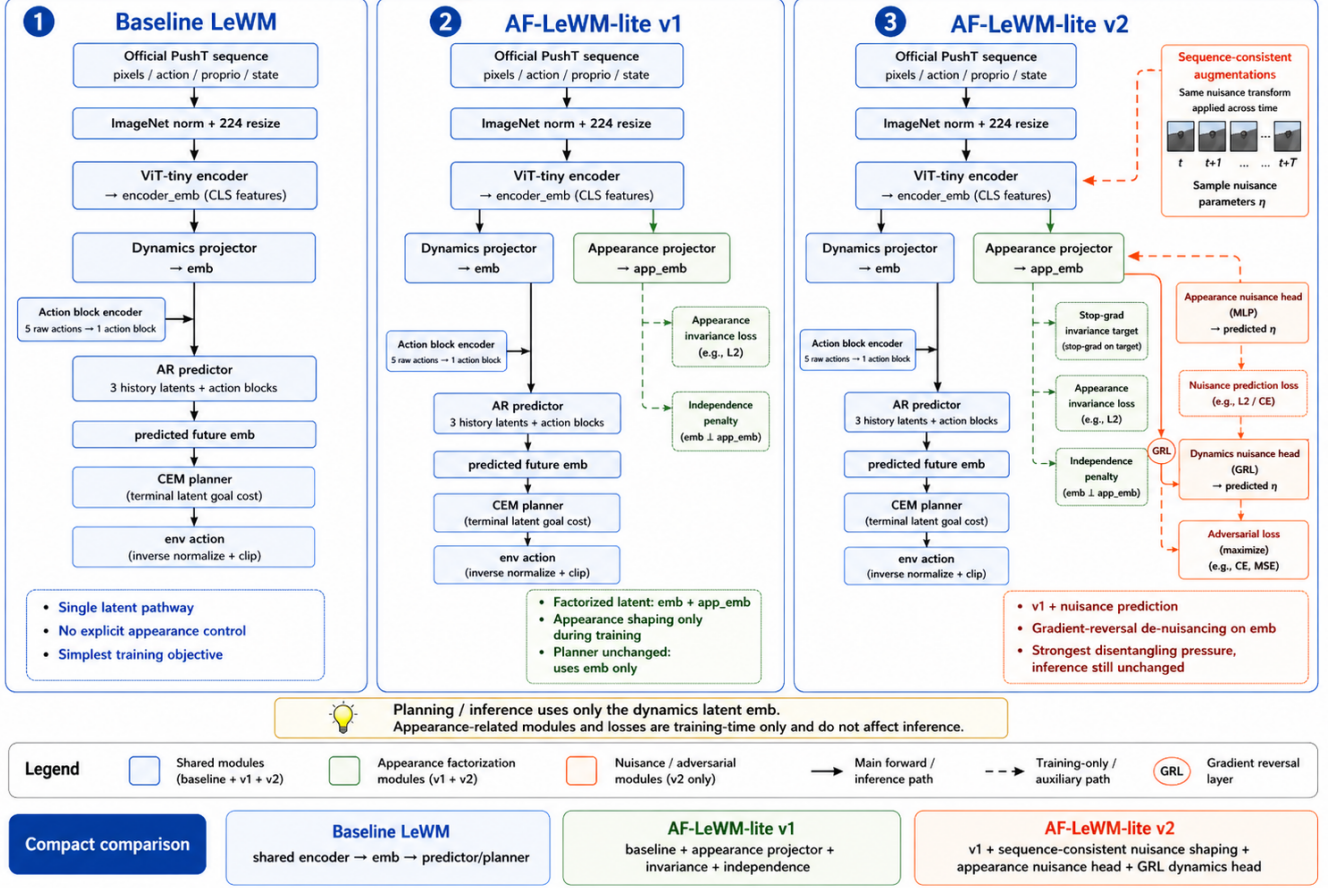


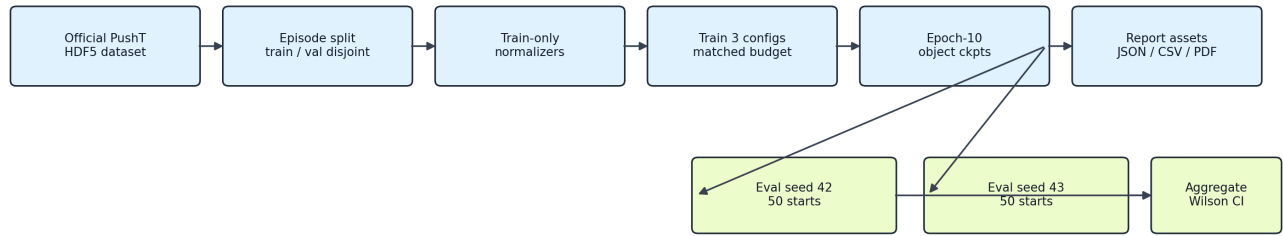
Figure 1: Model pipeline. The planner sees only the dynamics branch; appearance heads affect training losses.

Reliable Experiment Protocol

The rerun uses the official `pusht_expert_train.h5` dataset, an episode-disjoint train/validation split, normalizers fitted only on training episodes, and fresh-run guards that refuse to mix stale checkpoint or metrics artifacts with a new run.

Stage 1 screens every planned structure with train seed 3072, 10 epochs, eval seeds 42 and 43, and 50 PushT episodes per eval seed. Stage 2 scales the selected candidates baseline, v1_current, and v2_app_nuisance_only with train seeds 3072 and 3073, 50 epochs, eval seeds 42 through 45, and 50 episodes per eval seed. Stage 2 reports epoch-25 and epoch-50 checkpoints, giving 400 evaluation episodes per variant per checkpoint.

Reliable PushT Experiment Flow



Fresh-run fail-fast prevents stale checkpoint continuation.

Figure 2: Experiment flow from official dataset to checked aggregate reports.

Implementation Fixes

- Training now uses episode-disjoint subsets rather than overlapping clip-level random split.
- Evaluation reconstructs the exact training normalizers from the recorded train split metadata.
- Fresh reliable runs fail fast when old metrics or checkpoints already exist.
- Evaluation reconstructs the raw 15-step history window and compresses it to the model's 3-frame context.
- Goal indexing is asserted against the planned raw-step horizon.
- PushT goal rendering caches are synchronized after dataset-goal reset before replanning.
- Batched CEM evaluates all 50 envs together without expanding pixels or goal images over candidate samples.
- CEM candidate actions, elite means, and returned normalized plans are clamped to normalized env action bounds before inverse transform.
- Evaluation writes row, episode, start-step, train split, checkpoint, and raw dataset provenance.

Implementation Fix Map

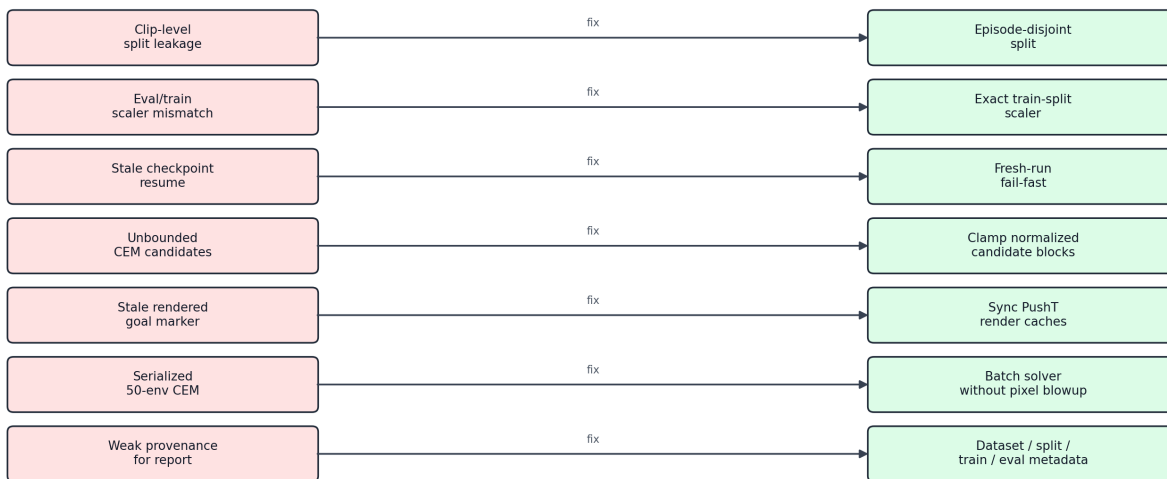


Figure 3: Reliability fixes that directly affect whether the experiment tests the intended design.

Results

Stage 1 variant	Family	Successes	Episodes	Success rate	Delta vs baseline
baseline	LeWM	6	100	6.0%	0.0 pp
v1_current	v1	10	100	10.0%	+4.0 pp
v1_inv_only	v1	2	100	2.0%	-4.0 pp
v1_indep_only	v1	5	100	5.0%	-1.0 pp
v1_seq_only	v1	5	100	5.0%	-1.0 pp
v1_seq_stopgrad	v1	3	100	3.0%	-3.0 pp
v2_app_nuisance_only	v2	8	100	8.0%	+2.0 pp
v2_weak_grl	v2	6	100	6.0%	0.0 pp
v2_current	v2	5	100	5.0%	-1.0 pp
v2_grl_warmup	v2	6	100	6.0%	0.0 pp

Table 1: Stage 1 structural screen. Two AF candidates beat baseline under the short 100-episode screen.

Checkpoint	Variant	Successes	Episodes	Success rate	Delta vs baseline
epoch 25	baseline	21	400	5.25%	0.00 pp
epoch 25	v1_current	19	400	4.75%	-0.50 pp
epoch 25	v2_app_nuisance_only	20	400	5.00%	-0.25 pp
epoch 50	baseline	25	400	6.25%	0.00 pp
epoch 50	v1_current	22	400	5.50%	-0.75 pp
epoch 50	v2_app_nuisance_only	20	400	5.00%	-1.25 pp

Table 2: Stage 2 reliability check. The scaled AF candidates trail baseline at both inspected checkpoints.

Checkpoint	Variant	Train seed	Successes	Episodes	Success rate
epoch 25	baseline	3072	11	200	5.50%
epoch 25	baseline	3073	10	200	5.00%
epoch 25	v1_current	3072	10	200	5.00%
epoch 25	v1_current	3073	9	200	4.50%
epoch 25	v2_app_nuisance_only	3072	8	200	4.00%
epoch 25	v2_app_nuisance_only	3073	12	200	6.00%
epoch 50	baseline	3072	14	200	7.00%
epoch 50	baseline	3073	11	200	5.50%
epoch 50	v1_current	3072	10	200	5.00%
epoch 50	v1_current	3073	12	200	6.00%
epoch 50	v2_app_nuisance_only	3072	13	200	6.50%
epoch 50	v2_app_nuisance_only	3073	7	200	3.50%

Table 3: Stage 2 per-training-seed detail.

Interpretation

The corrected result should be read conservatively. Stage 1 did not show that all AF variants were weaker than baseline: v1_current and v2_app_nuisance_only were ahead under the short screen. Stage 2 is the stronger reliability check, and both scaled AF candidates trail baseline there. The current study is useful as a reliability-checked ablation of the AF shaping idea; it does not establish an AF-LeWM-lite PushT improvement.

The main decision is to use baseline as the reliable PushT result for this repo. v1_current remains valuable as a negative mechanism result because it produces cleaner appearance/dynamics factorization diagnostics, but the cleaner split does not improve planning success under this budget.